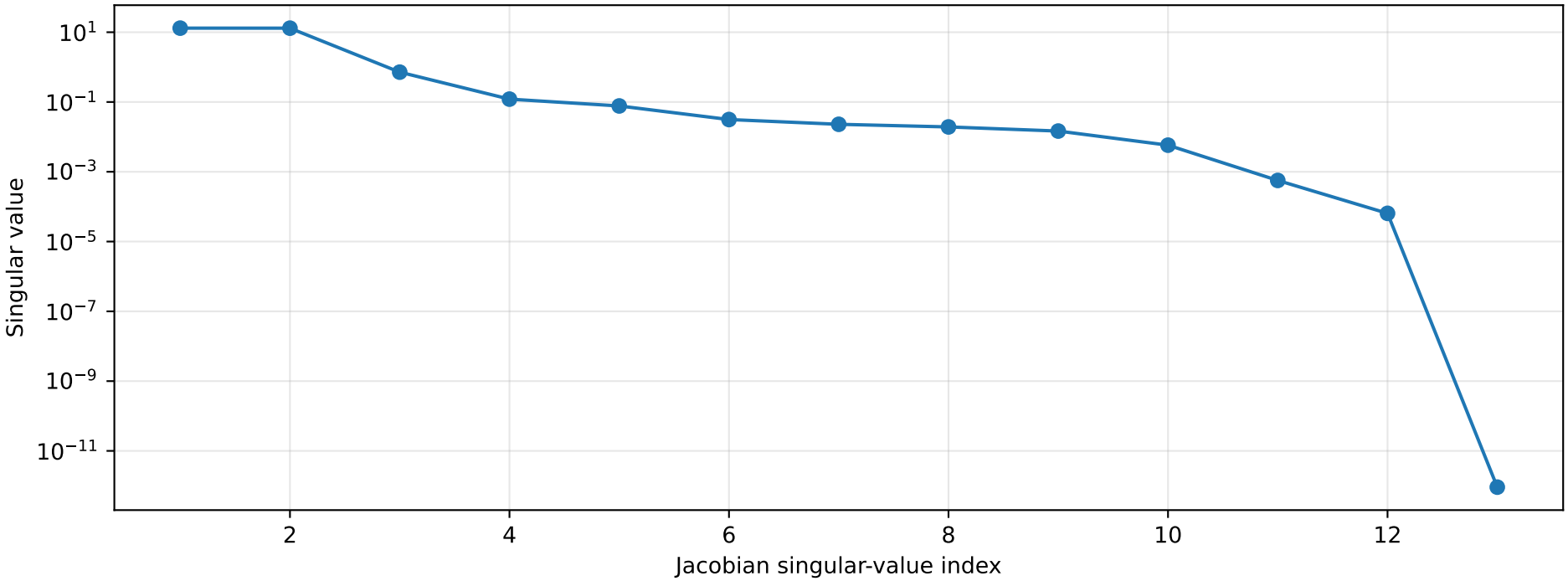
Mocap-refined nonlinear Euler model

Train: bag8 40-45 s; held-out prediction: 50.001-55.001 s
Centered SG 0.32 s / degree 3; normalized quaternion first/second derivatives
Ordinary time-weighted torque LS; nonlinear omega x J omega retained
Fixed nominal trace(J), rotor geometry and reaction ratio; 13 fitted coordinates

Parameter	Unit	Nominal / numerical start	Refined
Jxx	kg m2	0.06500006	0.1025516
Jyy	kg m2	0.06495266	0.02926971
Jzz	kg m2	0.1289921	0.1271235
Jxy	kg m2	-7.278993e-07	-0.0127573
Jxz	kg m2	1.901508e-05	0.0001084323
Jyz	kg m2	5.91673e-08	-0.01481887
CoG x	m	-0.002024709	0.04939255
CoG y	m	-3.052658e-05	0.01202008
CoG z	m	0.00950975	0.03041673
Rotor 0 thrust multiplier	1	1	0.3462125
Rotor 1 thrust multiplier	1	1	0.5300467
Rotor 2 thrust multiplier	1	1	0.6102281
Rotor 3 thrust multiplier	1	1	0.3475665
Thrust time constant	s	0.004983183	0.2442783

trace(J) = 0.2589448285 kg m2 (scale convention)
Log-chart rank 12/13; direct physical local rank 13/13
Near planar-inertia boundary: min eigenvalue(C)=2.8e-12
Rotational-only effective fit; translational force balance is not constrained.
Thrust tau seed is the median command period, not a physical nominal time constant.
Curvature covariance is unreliable near the boundary; SG errors are correlated.

Identification strength and held-out prediction



Hz	Readings	Nominal nonlinear	Refined nonlinear
0.5	3	103.5340	57.9916
2	10	51.4672	14.8143
5	25	22.3489	6.7866

Held-out SO(3) RMSE [deg], sigma=2.5 deg/s; same 501 score times.
Training residual mean is added and frozen before evaluation.
Same original gyro values at all noises; no synthetic draws.
Only sparse gyro and the initial mocap pose enter the predictor.
Bands condition on the fitted physical point, not unknown future model error.